

# AISEE-BIN: Flower Dome progress

Status update for the team · 1 October 2026 · Gardens by the Bay, Flower Dome POC



**The goal:** guide blind and low-vision visitors along the Flower Dome tour (Point A, through Doors B and C, to Points 1–4) with spoken directions. Guidance comes from a phone or from AiSee glasses.

## 1 Scan the place

Walk the route once with a phone so it can learn what the place looks like.

## 2 Add the route

In our web editor, draw the walkways and add the stops, hazards and what to say at each.

## 3 Guide the visitor

The phone or glasses recognise where they are and give spoken directions and announcements.

## Two ways to scan, and the best of both

### A · Our app scans and guides (ARKit)

The iPhone's own tracking, scanned in our app's Author mode

- + One walk gives the map, the route and the stops
- + Smooth, continuous tracking. **Worked inside the dome**
- iPhone only, and **not for glasses**
- **Struggled at the entrance**, and sensitive to light

### B · Immersal scans, our app guides

Scanned with the Immersal Mapper app; route added in our editor

- + **Found the entrance instantly**
- + Works for **phones and glasses**, on iOS and Android
- About 500 photos per map, so the full route needs several scans
- **Joining scans into one map is unsolved**

**Best of both:** Immersal tells the phone *where* it is, and ARKit tracks every step in between. The phone already does this on Immersal maps. Next we use it to extend maps and to join scans.

## Where we are

|                                              |                                                                                                                                      |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| ● <b>iPhone app</b> (TestFlight)             | Ready: navigation, scanning (Author mode) and on-site test tools.                                                                    |
| ● <b>Android app for the glasses</b>         | Ready as a test build. Smooth glasses video, and positioning works in a test room.                                                   |
| ● <b>Web map editor</b>                      | Live at <a href="https://aiseebin.flowsxr.com">aiseebin.flowsxr.com</a> . Anyone on the team can edit routes, stops and spoken text. |
| ● <b>Phone navigation at the Flower Dome</b> | Worked to Points 1–3 on 1 October. The entrance (A, B, C) needs a better map.                                                        |
| ● <b>Glasses at the Flower Dome</b>          | Not tested on site yet. Planned for <b>16 October</b> , with Prasanth in Singapore.                                                  |

## Timeline

| 16 Sep                                | 24–26 Sep                                             | 25 Sep                            | 29–30 Sep                                                    | 1 Oct              | 2 Oct                                 | by 16 Oct                              | 16 Oct               |
|---------------------------------------|-------------------------------------------------------|-----------------------------------|--------------------------------------------------------------|--------------------|---------------------------------------|----------------------------------------|----------------------|
| Glasses camera recognised by Immersal | Phone positioning in Immersal maps; on-phone matching | Immersal scans of the Flower Dome | Android glasses app, web editor upgrades, on-site test tools | First on-site test | Full-route Immersal scan, in daylight | Combined Immersal + ARKit; extend maps | Glasses test on site |

**Next:** on 2 October Yvonne scans the whole route as **one** Immersal map. If it holds together, phones and glasses can guide the full tour from a single map. Before 16 October we build the combined Immersal + ARKit positioning and the option to add to an existing map.

**Links:** [iPhone app \(TestFlight\)](#) [Android app \(APK\)](#) [Web map editor](#) [GitHub repo](#) [Immersal portal](#) [Immersal Mapper](#) · details on the last page

## 1 · The two ways in detail

|                          | A · Our app scans (ARKit)                                                                                                                                                                                                                    | B · Immersal scans, our app guides                                                                                                                                                                                                                       |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>How a map is made</b> | Open Author mode, name the place, walk the route, mark each stop, then Publish. The route comes out of the walk itself.                                                                                                                      | Scan with the Immersal Mapper app. Then <i>Import from Immersal</i> in our editor and draw the route and stops on top of the scan.                                                                                                                       |
| <b>How it finds you</b>  | The phone recognises the place from its saved world map (about 26 MB for the Flower Dome walk). It then tracks every step.                                                                                                                   | A camera frame is matched against the Immersal map and gives a position fix. Between fixes, the phone tracks with ARKit and the glasses count steps.                                                                                                     |
| <b>Devices</b>           | iPhone only.                                                                                                                                                                                                                                 | iPhone, Android and AiSee glasses.                                                                                                                                                                                                                       |
| <b>Strengths</b>         | <ul style="list-style-type: none"><li>• Quick to make: one walk.</li><li>• Very smooth tracking.</li><li>• Works offline once downloaded.</li><li>• Announcements arrived just before each point.</li></ul>                                  | <ul style="list-style-type: none"><li>• Recognises busy, repetitive places better (the entrance).</li><li>• Can match online or on the phone itself.</li><li>• The only option for the glasses.</li></ul>                                                |
| <b>Weaknesses</b>        | <ul style="list-style-type: none"><li>• Recognition fails with changing light or look-alike surfaces.</li><li>• Recognises best near where the scan started.</li><li>• Two scans can't be joined, and a map can't be extended yet.</li></ul> | <ul style="list-style-type: none"><li>• About 500 photos per map.</li><li>• A large site needs several scans, and lining them up by hand was out by tens of metres.</li><li>• The scanning and the route-making happen in two different tools.</li></ul> |

### Combining them

Immersal gives an occasional, absolute "you are here", and ARKit gives smooth, step-by-step tracking that drifts slowly. Together they give the strengths of both. Today the phone already uses ARKit between Immersal fixes. Three next steps:

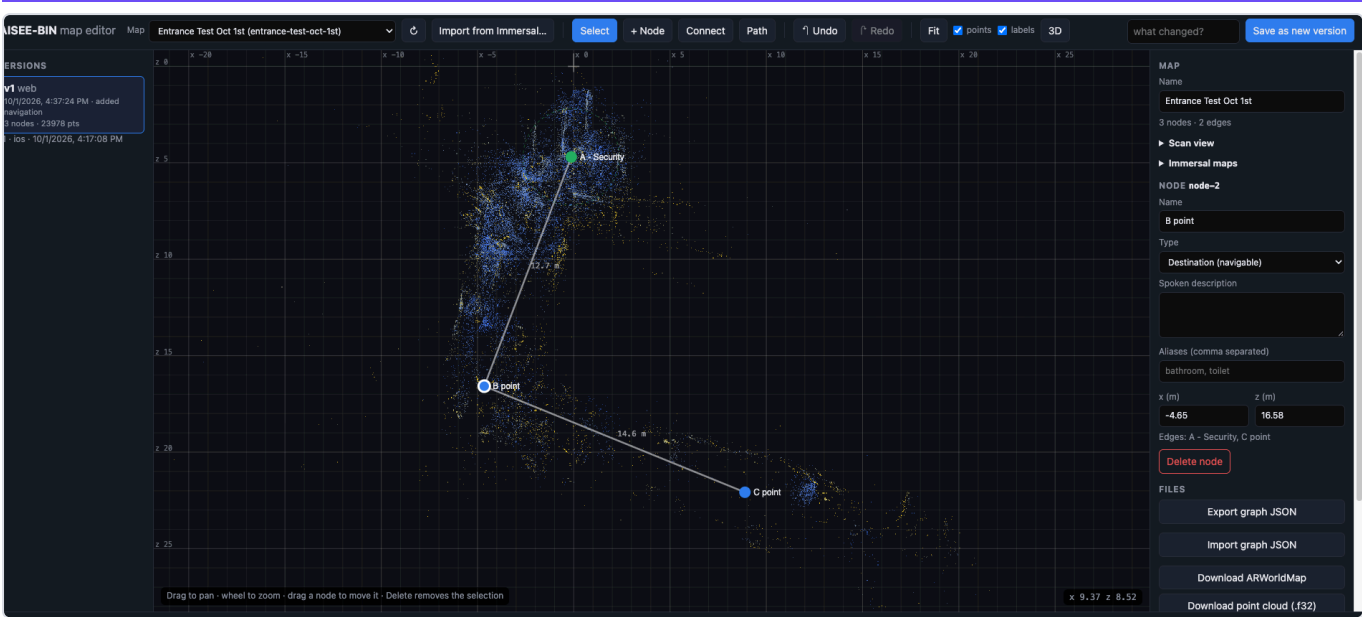
- **Add to an existing map.** Immersal places the phone in the map first, so a new walk lands in the same coordinates and extends the map instead of starting a new one.
- **Join Immersal scans.** A walk that sees two scans within a few seconds tells us exactly how they fit together. That is the "link scans" step of the on-site test; see section 4.
- **Author-mode Immersal maps.** Our Author mode can also send photos to Immersal as it scans (up to 150), so one walk gives both kinds of map. On 1 October no Immersal map came out of it; still to check.

## 2 · Matching online or on the device

With Immersal maps, a camera frame can be matched by Immersal's server or by the phone itself. The app has a switch with three settings: **Auto**, **On phone** and **Immersal server**.

|                 | Online (Immersal server)                                                                                                                                                    | On the device                                                                                                                                                                                                                                                                                             |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pros</b>     | <ul style="list-style-type: none"><li>• Nothing to download.</li><li>• Always uses the latest map.</li><li>• Light on the phone's processor and battery.</li></ul>          | <ul style="list-style-type: none"><li>• Works with no internet. Glasses use their own Wi-Fi, so the phone has no internet while streaming unless it has mobile data.</li><li>• Fast with one map: about 0.3 s per fix on our Android test phone.</li><li>• Camera frames never leave the phone.</li></ul> |
| <b>Cons</b>     | <ul style="list-style-type: none"><li>• Needs mobile data.</li><li>• About 1–3 s per try on site, sometimes up to 8 s.</li><li>• Requests timed out on 1 October.</li></ul> | <ul style="list-style-type: none"><li>• The map must be downloaded first.</li><li>• Slower when several maps are loaded: 1.5–8 s per try with the four Flower Dome scans on an iPhone 13 Pro Max.</li><li>• Slower on older phones.</li><li>• One iPhone issue still to fix.</li></ul>                    |
| <b>Best for</b> | Phones with good mobile data.                                                                                                                                               | The glasses, and places with poor signal. Ideally one map per area.                                                                                                                                                                                                                                       |

### 3 · The web map editor (aiseebin.flowsxr.com)



The editor with last week's Immersal entrance scan. A, B and C have been added, and stop "B point" is selected; its settings are on the right.

#### What you can do

- **Open any map** from the Map menu. Each saved change becomes a new **version**, listed on the left, so nothing is lost.
- **Import from Immersal** to start a map from an Immersal scan.
- See the scan as a **point cloud**, top-down or in **3D**, to tell where walls, paths and beds are.
- **Line up several Immersal scans** into one map (Immersal maps panel).
- Download the map files (route data, world map, point cloud).

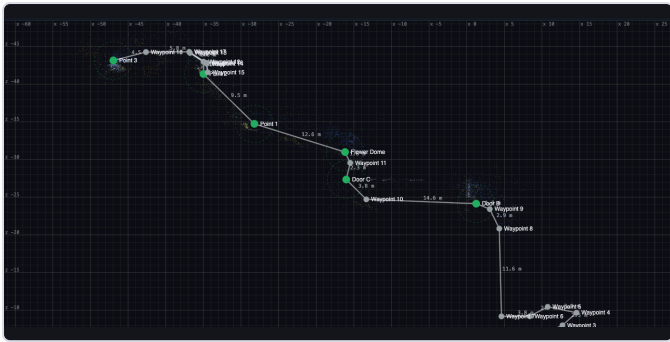
#### How to add or edit a stop

1. **+ Node** and click on the map; drag to move it.
2. Set the **Type**:
  - *Exhibit*: a stop that is announced in passing.
  - *Hazard*: a warning, with a vibration.
  - *Destination*: a place visitors can ask to go to.
  - *Junction*: a routing point only.
3. Write the **Spoken description**, add **Aliases** (other names visitors might say), and set how close to **announce** it.
4. **Connect** nodes (or draw a **Path**) to make walkways.
5. Type what changed and click **Save as new version**. The apps pick it up the next time they open the map.

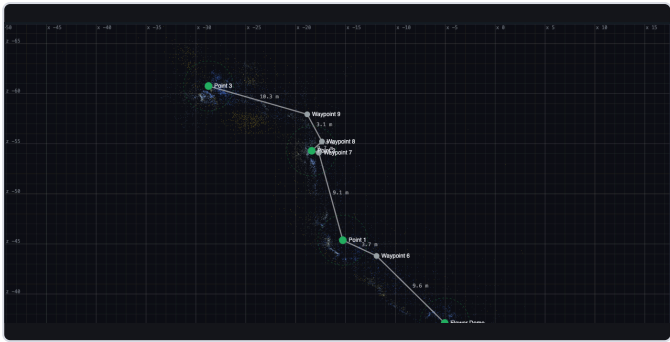
### 4 · What we tested on site on 1 October

Our tester, on an iPhone 13 Pro Max, with the TestFlight app and remote support. Her logs and published maps reached our server, so we could follow along from overseas.

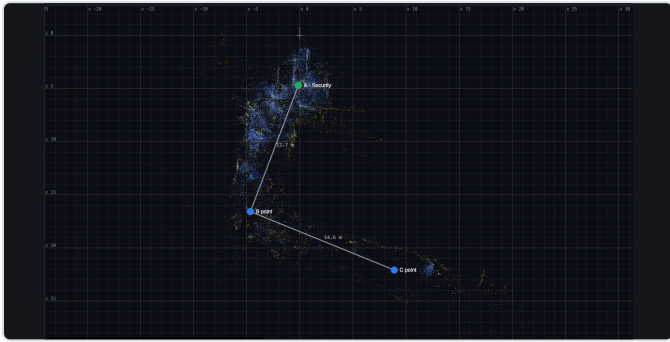
| Test                                                                                      | What happened                                                                                                       | Result                                                                                                                                |
|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Five Immersal scans as one map</b><br>last week's scans, lined up by hand              | She walked to link the scans, marked the tour points, then checked them. Linking timed out at first on mobile data. | <b>Not usable yet.</b> Checks were 13–47 m off. The linking walk gave a better fit, but it still disagrees with her marked points.    |
| <b>Way A: "Flower Dome 1 Oct"</b><br>about 108 m, Point A to Point 3                      | Published after a couple of failed uploads. She then navigated to Points 1, 2 and 3 and back.                       | <b>Worked well inside the dome.</b> Each announcement came just before its point.<br><b>It had trouble finding her at A, B and C.</b> |
| <b>Way A: "map 2"</b><br>the actual priority-lane route                                   | Retested later, as it was getting darker.                                                                           | <b>Stuck on "Relocalizing"</b> at the entrance. Likely causes are the change in light and the barrier graphics.                       |
| <b>Way B: "Entrance Test Oct 1st"</b><br>Immersal entrance scan, A, B and C placed by eye | Loaded on the phone at the entrance.                                                                                | <b>Found her immediately</b> , at roughly the right place. The points are only rough because they were placed by eye.                 |



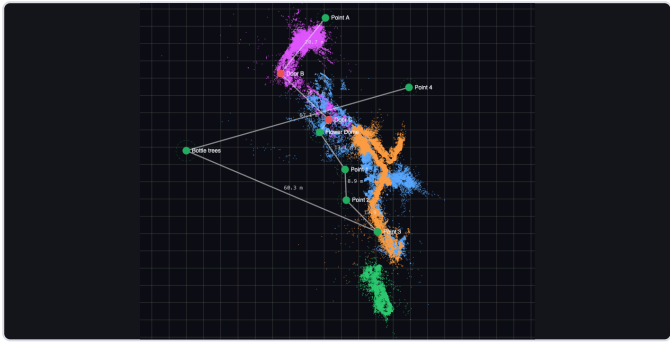
Way A, "Flower Dome 1 Oct": Point A through Doors B and C to Points 1–3, from one walk.



Way A, "map 2": the priority-lane route. It got stuck relocalizing at the entrance.



Way B, "Entrance Test Oct 1st": the Immersal entrance scan. It found her position immediately.



The five Immersal scans (one colour each), lined up by hand. They look plausible, but were 13–47 m off on site.

## 5 · Glasses: tested so far

The glasses have no ARKit, so they rely on Immersal maps (Way B). The phone stays in a pocket and does the work.

| When                | What we found                                                                                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16 Sep · iPhone     | Immersal recognises the AiSee glasses' camera picture. We worked out the camera's lens setting (about a 60° view).                                                                                     |
| 24 Sep · iPhone     | First walking test with the glasses. Few fixes came through (3 in 474 tries), so this needs work.                                                                                                      |
| 29 Sep · iPhone     | New glasses software from the vendor. The camera preview lag turned out to be on our side; with that fixed, the video is now smooth.                                                                   |
| 29–30 Sep · Android | A glasses-only Android app on a Huawei Nova 5T, a phone without ARCore. In a test room: 25 fixes in 168 tries, about 0.3 s each, all on the phone. Spoken announcements while walking, and a live map. |
| 16 Oct · on site    | <b>First glasses test in the Flower Dome</b> , with Prasanth in Singapore, using the Immersal map of the route. Controls: tap the temple to talk, double-tap for "where am I", triple-tap to stop.     |

## 6 · What we learned

- **Light matters for both ways.** Tours run in the daytime, so scans should be made in daylight.
- **The entrance is the hard part.** It has look-alike surfaces, graphics on the barriers and changing light. Immersal handled it better than ARKit.
- **One map is far easier than many.** Lining up scans by hand was out by tens of metres, and automatic matching is unreliable on plants. Scanning the whole route as one Immersal map is the quickest thing to try.
- **Remote testing works.** Field Test logs and the editor let us diagnose problems while our tester was on site.

## 7 · Next steps

| When  | What                                                                                                                                                                                                                    |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2 Oct | Yvonne makes one Immersal scan of the whole route within the 500-photo limit, split by area (100+ photos at the entrance), in daylight. She also sends a screen recording and test results; we add them to this report. |

- |               |                                                                                                                                                                                                                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Before 16 Oct | <ul style="list-style-type: none"><li>• Combined Immersal + ARKit positioning.</li><li>• "Add to an existing map".</li><li>• Fix the iPhone on-device issue, then settle how the five scans fit together.</li><li>• Faster, more accurate glasses positioning.</li><li>• Write the spoken cues into the final map.</li></ul> |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|        |                                                                |
|--------|----------------------------------------------------------------|
| 16 Oct | On site in Singapore: glasses and phone test of the full tour. |
|--------|----------------------------------------------------------------|

## 8 · Links

| What                                     | Link                                                                                                     | Notes                                                                       |
|------------------------------------------|----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| <b>iPhone app</b> (TestFlight)           | <a href="https://testflight.apple.com/join/7uVqcqkE">testflight.apple.com/join/7uVqcqkE</a>              | Install TestFlight first, then open the link. Build 10.                     |
| <b>Android app</b> (glasses only)        | <a href="#">AISEE-BIN-android-0.1.1.apk</a><br>release page: <a href="#">releases/tag/android-v0.1.1</a> | Needs access to the GitHub repo. Allow "install unknown apps" on the phone. |
| <b>Web map editor</b>                    | <a href="https://aiseebin.flowsxr.com">aiseebin.flowsxr.com</a>                                          | Every save becomes a new version, so edits can always be undone.            |
| <b>Code and documents</b>                | <a href="https://github.com/prasanthsasikumar/AISEE-BIN">github.com/prasanthsasikumar/AISEE-BIN</a>      | Private repo: ask Prasanth for access. Reports are in docs/reports/.        |
| <b>Immersal developer portal</b>         | <a href="https://developers.immersal.com">developers.immersal.com</a>                                    | Where our Immersal maps are listed and managed (Pro account).               |
| <b>Immersal Mapper</b><br>(scanning app) | <a href="#">App Store: Immersal Mapper</a>                                                               | Sign in with our Immersal Pro account, so the app can use the scans.        |
| <b>On-site tester guide</b>              | <a href="#">docs/reports/2026-10-01-field-test-guide</a>                                                 | Step-by-step guide to the Field Test tools.                                 |